

Vision & Immersion Working Group
De Vinci Research Center
Contact: Ahmed.azough@devinci.fr

The research subjects can be related to the technologies of:

- Computer vision
- Immersive technologies
- Digital twins

If you are interested in joining this group, please send me your confirmation. Please forward this email on to any DVRC colleagues who may be interested in joining the group.

Here are some possible research areas that could be held within this group:

- Immersive technologies
 1. Human-Computer Interactions and feedback
 2. Augmented, Virtual and Mixed Reality experiences
 3. Immersion, presence, engagement and collaboration in immersive environments
 4. Multimodal interactions in extended realities: vision, sound, haptic, etc.
 5. AI & Metaverse, Realistic and responsive virtual characters
 6. Security, privacy and ethics in Virtual environments
 7. Networks and Connectivity for more realistic social immersion
 8. Emerging economic models in virtual worlds
 9. Accessibility and inclusion in the Metaverse
 10. Metaverse and virtual education to enhance teaching and training methods
 11. Immersive technologies for health and well being
 12. Content Creation and Virtual Design
- Computer vision
 1. Object Recognition and Classification and Image Segmentation
 2. Facial Recognition, emotion detection and object re-identification
 3. Gesture recognition and Human Pose Estimation
 4. Actions, activities and event recognition
 5. Deep Learning and Generative AI
 6. 3D scene understanding from 2D images and videos
 7. Robotic Vision and Visual Simultaneous Localization and Mapping (SLAM) and
 8. Medical Image Analysis and interpretation
 9. Autonomous Vehicles for autonomous navigation
 10. Generation, compression, streaming and Deepfake
- Digital twins:
 1. Lifecycle Management and Interoperability
 2. Applications in Industry, Infrastructure, Smart Cities, and Blockchain
 3. Modeling and Simulation
 4. Data Integration and Fusion
 5. Real-Time Monitoring and Analytics

As we're still at the initiation stage, these are all proposals, so don't hesitate to suggest changes or additions of any features you think are important.

